University of Oxford
St Edmund Hall

July 2026

LETTER OF ACCREDITATION

Dear Registrar,

ZIHE ZHOU

We are pleased to write to you with the transcript and Letter of Accreditation for the above student, who was studying at Oxford as a non-graduating Visiting Student at St Edmund Hall.

A full explanation of the grades earned is included. We provide all Visiting Students with copies of this report file and we have no objection to it becoming part of the student's academic file.

Visiting Student is a status recognised and conferred by St Edmund Hall, one of the 38 self-governing colleges within the University of Oxford, and by the University of Oxford itself.

Visiting Students are either one-, two-, or three-term students, normally on leave from their home college where they are degree candidates. They are not candidates for Oxford degrees, nor do they matriculate but are registered with the University of Oxford as Visiting Students. Visiting Students are taught by the same specialist tutors who teach Oxford degree candidates; they are taught to the same standard and using the same methods.

Visiting Students also have full access to the College and University Libraries, and enjoy full Junior Common Room social facilities and dining rights in Hall. Graduate Visiting Students also become members of the Middle Common Room. Visiting Students are encouraged to represent their College in sporting and cultural activities.

In short, Visiting Students are offered, as fully as possible, the academic and self-improving opportunities of life at Oxford University. The work undertaken by a Visiting Student is entirely arranged, supervised and evaluated by St Edmund Hall, and is recommended for credit-transfer, where appropriate, to your institution.



College Registrar
St Edmund Hall

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Registered Charity № 1137470

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EXPLANATION OF THE TRANSCRIPT GRADING AND RECOMMENDED CONVERSION SCALE TO ALPHABETICAL GRADES

St Edmund Hall Visiting Students are taught (generally singly or in pairs) and given termly assessments by their tutors. The approximate conversion scale below is provided for the interest of completed Visiting Students, and recommended to the Registrars and Faculty of interested academic bodies who may seek to transfer grades as well as credit.

Please note that the final translation into a grade is entirely up to the home institution's judgement, and the home institution is not bound by the approximate guidance below.

The grades for termly work given by tutors to Oxford's own undergraduates do not themselves determine the class of degree the students will eventually earn. The degree classifications are based on examination papers (Finals) set usually at the end of their courses. Nevertheless the termly grades constitute regular, and usually highly accurate, assessments of the students' likely performance in the Final Examination. Tutors and students alike are entirely accustomed to this grading system. Visiting Students, not being candidates for Oxford degrees, do not sit Oxford Finals. However, they are still subject to termly reports and their tutors therefore made grade assessments according to the scales with which they are familiar.

In Oxford, lectures are offered during an intensive 8-week period in each of 3 terms during the academic year, but almost all students will need to do academic work for at least 10 weeks each term, including the planning of tutorials. The work done for an Oxford tutorial is usually regarded as equivalent to the work done for one semester credit.

The academic standard at Oxford is high, and only students of high ability are accepted as Visiting Students by St Edmund Hall. It is unlikely that the course of any Visiting Student performing consistently at or below the base grades set out on the Conversion Scale would be prolonged.

Recommended Conversion Scale

Degree Class	EPSC Exam Mark Scale	Alphabetical Scale (approx. equivalent)
I	80-100	A+
	70-79	A
II (i)	65-69	A-
	60-64	B+
II (ii)	55-59	B
	50-54	B-
III	45-49	C+
	40-44	C
Pass	30-39	C-
Fail	0-29	Fail – no Grade

University of Oxford
St Edmund Hall

ACADEMIC REPORT

Academic Year 2025-26

This is to certify that: Zihe Zhou **DoB.** 02/06/2005 **Student No.** 1846086

was enrolled as a Visiting Student of St Edmund Hall and has completed those tutorial courses indicated below. Please refer to the ***Explanation of the Transcript Grading and suggested Conversion Scale to American grades***.

COURSES STUDIED	PERSONAL TUTOR	OXFORD GRADE	NUMBER OF TUTORIALS
Michaelmas Term 2025 (October - December)			
Quantum Theory	Dr T Crawford	78	4
B8.4 Information Theory	Departmental classes	85	3 classes
B5.2 Applied Partial Differential Equations	Departmental classes	67	3 classes
Hilary Term 2026 (January - March)			
C7.4 Introductions to Quantum Information	Departmental classes	67	3 classes
C8.2 Stochastic Analysis and PDEs	Departmental classes	75	4 classes
B7.3 Further Quantum Theory	Departmental classes	69	4 classes
Trinity Term 2026 (April - June)			
Graph Theory	Prof O Riordan	75	4
Number Theory	Mr J Ait Idir La Huerta	70	4
Calculus of Variations	Mr B Ahmadpour	100	4

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Visiting Student Report Form

Michaelmas Term 2025

Name of student:	Zihe Zhou
Name of tutor:	Dr T Crawford
Number of tutorials:	4

Mark:	78	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **Quantum Theory**

Quantum theory was born out of the attempt to understand the interactions between radiation, described by Maxwell's theory of electromagnetism, and matter, described by Newton's mechanics.

Although there remain deep mathematical and physical questions at the frontiers of the subject, the resulting theory encompasses not just the mechanical but also the electrical and chemical properties of matter. Many of the key components of modern technology such as transistors and lasers were developed using quantum theory.

In quantum theory particles also have some wave-like properties. This introductory course explores some of the consequences of this culminating in a treatment of the hydrogen atom.

By the end of this course, students will be able to solve the Schroedinger equation in a range of simple situations and understand its significance. In particular they will be able to calculate the energy levels of hydrogen-like atoms. They will also learn the abstract, algebraic formulation of quantum mechanics which complements and sometimes replaces the solution of the Schroedinger equation.

REPORT:

Zihe has submitted excellent work throughout the Quantum course. She is very responsive in the tutorials and offers insightful contributions to the discussion. I have also enjoyed her enthusiasm for the course material and her enjoyment of learning. A real pleasure to teach.

Achievement - Excellent

Date: 15/01/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Michaelmas Term 2025

Name of student:	Zihe Zhou
Name of tutor:	Departmental classes
Number of tutorials:	3 classes

Mark:	85	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **B8.4 Information Theory**

Information theory is a relatively young subject. It played an important role in the rise of the current information/digital/computer age and still motivates much research in diverse fields such as statistics and machine learning, physics, computer science and engineering. Every time you make a phone call, store a file on your computer, query an internet search engine, watch a DVD, stream a movie, listen to a CD or mp3 file, etc., algorithms run that are based on topics we discuss in this course. However, independent of such applications, the underlying mathematical objects arise naturally as soon as one starts to think about "information" in a mathematically rigorous way. In fact, a large part of the course deals with two fundamental questions: How much information is contained in a signal/data/message, and how do we compress it? (source coding).

What are the limits to information transfer over a channel that is subject to noisy perturbations, and how to encode and decode in this setting? (channel coding)

REPORT:

Submitted consistently excellent work on all problem sets, with only very minor errors.

Date: 11/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Michaelmas Term 2025

Name of student:	Zihe Zhou
Name of tutor:	Departmental Classes
Number of tutorials:	3 classes

Mark:	67	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **B5.2 Applied Partial Differential Equations**

This course continues the Part A Differential Equations courses. In particular, first-order conservation laws are solved and the idea of a shock is introduced; general nonlinear and quasi-linear first-order partial differential equations are solved, the classification of second-order partial differential equations is extended to systems, with hyperbolic systems being solved by characteristic variables. Then Riemann's function, Green's function and similarity variable methods are demonstrated.

REPORT:

Excellent work across all problem sets - well done.

Date: 11/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Hilary Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Departmental Classes
Number of tutorials:	3 classes

Mark:	67	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **C7.4 Introductions to Quantum Information**

The classical theory of computation usually does not refer to physics. Pioneers such as Turing, Church, Post and Goedel managed to capture the correct classical theory by intuition alone and, as a result, it is often falsely assumed that its foundations are self-evident and purely abstract. They are not! Computers are physical objects and computation is a physical process. Hence when we improve our knowledge about physical reality, we may also gain new means of improving our knowledge of computation. From this perspective it should not be very surprising that the discovery of quantum mechanics has changed our understanding of the nature of computation. In this series of lectures you will learn how inherently quantum phenomena, such as quantum interference and quantum entanglement, can make information processing more efficient and more secure, even in the presence of noise.

REPORT:

Solid work with mostly correct solutions.

Date: 11/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Hilary Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Departmental Classes
Number of tutorials:	4 classes

Mark:	75	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **C8.2 Stochastic Analysis and PDEs**

Stochastic processes - random phenomena evolving in time - are encountered in many disciplines from biology, physics, geology through to economics and finance. This course focuses on developing the mathematics needed to describe stochastic processes evolving continuously in time and introduces the basic tools of stochastic calculus which are a cornerstone of modern probability theory. The canonical example of such a stochastic process is Brownian motion, also called the Wiener process. This mathematical object was initially proposed by Bachelier, as a model for asset prices, and by Einstein to describe the displacement of a pollen particle in a fluid. The paths of Brownian motion, or of any continuous martingale, are of infinite variation (they are in fact nowhere differentiable and have non-zero quadratic variation) and one of the aims of the course is to define a theory of integration along such paths as well as a suitable version of the chain rule, given by Itô's formula.

REPORT:

Perfect attendance and strong submissions to the problem sheets.

Date: 11/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Hilary Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Departmental Classes
Number of tutorials:	4 classes

Mark:	69	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **B7.3 Further Quantum Theory**

This course builds on Part A Quantum Theory. The mathematical foundations of quantum theory are developed more deeply than in the Part A course, and general principles regarding the realisation of symmetries in quantum mechanical systems are developed. Systems of several particles are studied, including a consideration of identical particles and particle statistics. Along the way, a number of simple-but-relevant concepts from the theories of Lie groups, representation theory, and functional analysis are introduced in a hands-on fashion. In semi-realistic systems, an exact solution to the quantum dynamics is rarely forthcoming. The second part of the course is largely devoted to developing several approximation techniques that can be applied in these more general settings. These are employed to study problems such as the determination of energy levels of the Helium atom. The course concludes with a short introduction to scattering theory in one spatial dimension.

REPORT:

Good concise calculations but at times your working is somewhat unclear and some parts are lacking in discussion and interpretation. It is clear you have good understanding of the material so please show the stages of your thought process in solutions.

Date: 11/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Trinity Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Prof O Riordan
Number of tutorials:	4

Mark:	75	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **Graph Theory**

General introduction to graph theory. Basic properties of paths, walks, cycles and trees. Euler circuits, Hamiltonian cycles and other long cycles in graphs. Hall's Theorem. Application and analysis of algorithms for minimum cost spanning trees, shortest paths and bipartite matching. Cayley's formula. Vertex colourings of graphs, including the chromatic polynomial and Brooks' Theorem.

REPORT:

We based the tutorials on the two problem sheets for the Part A Graph Theory course, and then the first two sheets from the Part B Graph Theory course. On the first sheet, Zihe's work showed good insight, but there were a lot of missing details in her arguments. We discussed this, and from there things improved a lot. This subject is actually quite a difficult one to formulate rigorous arguments in, since the ideas themselves are so visual and intuitive, but Zihe picked this up very well, and was producing some really good work by the end. Overall she was clearly working at 1st class level by our standards for Oxford students, which are very high. It was a pleasure teaching her, and I hope she continues to work in areas related to graph theory/combinatorics.

Achievement – Excellent

Date: 12/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Trinity Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Mr J Ait Idir La Huerta
Number of tutorials:	4

Mark:	70	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **Number Theory**

Number theory is one of the oldest parts of mathematics. For well over two thousand years it has attracted professional and amateur mathematicians alike. Although notoriously `pure' it has turned out to have more and more applications as new subjects and new technologies have developed. Our aim in this course is to introduce students to some classical and important basic ideas of the subject. Students will learn some of the foundational results in the theory of numbers due to mathematicians such as Fermat, Euler and Gauss. They will also study a modern application of this ancient part of mathematics.

REPORT:

Zihe showed great understanding of the subject and a lot of mathematical maturity when tackling the problems. Most of the sheets were very well and she was able to understand complex ideas quickly and use them to solve new exercises.

Achievement - Excellent

Date: 04/06/2026

University of Oxford
St Edmund Hall

Visiting Student Report Form

Trinity Term 2026

Name of student:	Zihe Zhou
Name of tutor:	Mr B Ahmadpour
Number of tutorials:	4

Mark:	100	TICK IF FINAL	/

(Please use only the numerical mark scale in line with the accompanying Guidance Sheet)

COURSE DESCRIPTION: **Calculus of Variations**

This course introduces variational methods and optimisation techniques for functionals, including Euler–Lagrange equations, constrained optimisation, variational principles in physics, and applications to mechanics and differential equations. The course focuses on both theoretical foundations and practical problem-solving approaches.

REPORT:

The student demonstrated outstanding performance. I would award an overall mark of 100 and the highest possible distinction (A+).

Achievement - Excellent

Date: 21/07/2026